

BVC24 · BHARATH VENDING CORPORATION

BVC24 Manufacturing ERP

Complete System Documentation · Edition 1.0

Engineering & Implementation Reference

Chennai, Tamil Nadu, India
www.bvc24.in

BVC24 ERP — Complete System Documentation

Project: BVC24 — Vendor-based Manufacturing ERP **Customer:** Bharath Vending Corporation (BVC)
Edition: 1.0 — Production Ready **Document Class:** Engineering & Implementation Reference **Format:** Master single-file edition

How to Read This Document

This is the master single-file edition of the BVC24 documentation. It is suitable for:

- Sending as a single PDF to a customer or stakeholder.
- Printing to A4 for offline review.
- Hand-off to a CTO, architect, or development team during a transition.

Each section corresponds to a focused file in the multi-file `docs/` tree — see `docs/README.md` for that structured navigation. Cross-references between sections inside this master file use anchor links.

Part I — Foundation

1. Executive Summary

1.1 Product Identity

BVC24 is a vendor-based manufacturing ERP designed for **Bharath Vending Corporation (BVC)** — a Chennai-headquartered manufacturer of automated vending machines (snack, beverage, combo, medicine, and special-purpose dispensers).

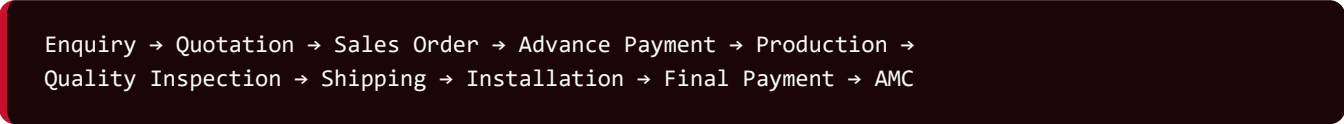
The platform consolidates the day-to-day operations of a manufacturing-and-installation business into a single web application:

- **Sales** — enquiries, quotations, sales orders, payment milestones.
- **Production** — product catalog, bill of materials, work orders, process stages, Gantt scheduling, quality inspections.
- **Procurement** — supplier master, purchase orders, goods receipt notes, inventory updates.
- **People** — employees, attendance (biometric), leave with email-token approvals, payroll, monthly STAR performance ratings.

- **Customer Lifecycle** — 360° customer view, requirements pipeline, post-sale installation projects.
- **Operations** — daily task allocation, real-time dashboards, MD-level WhatsApp alerts.

1.2 Business Value Proposition

BVC24 replaces a patchwork of spreadsheets, WhatsApp chats, and paper trails with a single source of truth that follows the cash flow:



Every state transition is timestamped, logged in an activity timeline, and gated by a business rule:

- A **Sales Order cannot enter production** until the customer's advance payment is recorded.
- A **Work Order cannot be closed** until its QC inspection is finalised.
- A **GRN feeds inventory only after finalisation** — preventing accidental stock changes.
- **Leave requests issue a single email** with one-click tokens to the approver — no portal login required.

These guard rails turn process discipline into a property of the software, not human memory.

1.3 Stakeholder Map

Stakeholder	Primary Screens	Key Outcomes
Managing Director	MD Review, Dashboard, WhatsApp alerts	Cash-in events, top performers, blocked SOs
Sales Team	CRM, Quotations, Sales Orders	Quote → SO conversion, payment progress
Production Head	Production, Work Orders, Gantt	Stage progress, employee allocation
Procurement	Purchase Orders, Suppliers, Inventory	PO funnel, GRN, supplier performance
Quality	Quality Management, NCRs	Inspection pass-rate, rework
HR	Employees, Attendance, Leave, Payroll	Compliance, payouts, recognitions
Floor Employees	Employee Dashboard, Apply Leave, Biometric	Today's tasks, leave balance
Customer (external)	Public Quotation link /q/:token	View & respond without an account

1.4 Scope of Implementation

Implemented and live (Release 1.0):

-  Phase 1 — Organization, IAM, Employees (profile, biometric, photos)
-  Phase 2 — Attendance, Leave (email-token approval), Payroll, STAR Performance
-  Phase 3 — CRM, Quotations with public share token, auto-pricing from BOM
-  Phase 4 — Procurement (Suppliers, POs, GRN with partial-receipt support)
-  Phase 5 — Sales Orders with payment-gated workflow, auto-project spawning
-  Production — Models, BOM, Work Orders, Process Stages, Gantt
-  Quality — Inspection checklists, NCRs, finalisation gate
-  Machines — Auto-registered from Work Order completion
-  Dashboard & Analytics — real-time stats, voice alerts
-  Notifications — Email (Resend + SMTP), WhatsApp (CallMeBot + Cloud API), in-app, voice
-  Chatbot — HR Assistant (rule-based), general ERP chatbot (Gemini)

Planned (not in this release):

-  Phase 6 — Invoicing, GST e-Invoice, e-Way Bill
-  Phase 7 — Service / AMC, complaint tickets

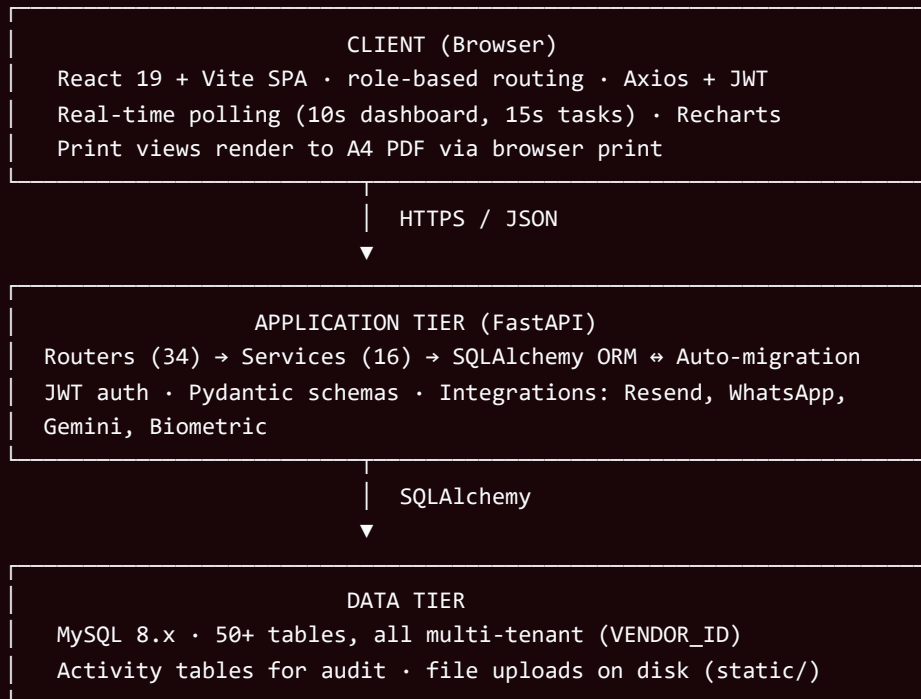
1.5 Headline Metrics

Surface	Count
Backend route files	34
HTTP endpoints	~200
Frontend pages	35
Print views	4
Public (no-auth) routes	2
Database tables	50+
Third-party integrations	5 (Resend, SMTP, Gemini, CallMeBot, WhatsApp)

2. System Architecture

2.1 High-Level View

A standard three-tier web application:



2.2 Backend Layered Structure

```
backend/app/
├── main.py                ← FastAPI app, CORS, router registration,
                           auto-migration on startup
├── database/database.py   ← engine, SessionLocal, get_db
├── auth/                  ← JWT bearer + encode/decode
├── models/models.py       ← All SQLAlchemy ORM models
├── schemas/               ← Pydantic schemas (22 files)
├── routes/                ← HTTP endpoint routers (34 files)
└── services/              ← business logic (16 files)
```

- **Routers** validate inputs, call services or the DB, shape the JSON response.
- **Services** encapsulate cross-cutting business logic (pricing, email, WhatsApp, payroll computation).
- **Models** describe every table; the same ORM class serves both persistence and business identity.
- **Schemas** are Pydantic v2 classes used for request validation and response serialisation.

2.3 Frontend Layered Structure

```
frontend/src/
├── main.jsx               ← React mount + BrowserRouter
├── App.jsx                ← Routes, role-based landing
├── App.css                ← Global theme (BVC red palette)
├── services/api.js        ← Axios instance, JWT injection
├── pages/                 ← 35 screens
└── components/            ← shared UI (ChatBot, HRAssistant, EntityDrawer, etc.)
```

- **No external UI library** — all UI is custom CSS-in-JS with inline styles + a small CSS file.
- **BVC red palette:** `#C8102E` primary, `#8B0B1F` dark, `#4A0E18` wine, `#1A0508` wine-black, `#F4B324` gold.
- **Sticky-header modal pattern** is reused across every detail modal in the system.
- **Real-time polling** every 10s (dashboard) / 15s (tasks). No WebSockets in this release.

2.4 Multi-tenancy

Every customer-facing table carries a `VENDOR_ID` referencing `vendor.ID`. The JWT carries `vendor_id` from login; every list query filters by it. Cross-tenant access is impossible without forging the token.

Currently BVC operates as `VENDOR_ID = 1`. Onboarding additional tenants requires no schema changes.

2.5 Worked Example — Sale to Shipment

1. Customer Apollo Hospitals enquires
 - POST /create-customer → Customer (LEAD_STATUS=NEW)
 - POST /customers/enquiry → WhatsApp MD alert
 - POST /customers/{id}/requirements
2. Quotation
 - POST /quotations/from-requirements → auto-priced from BOM
 - POST /quotations/{id}/send → STATUS=SENT, email + public token
3. Customer approves (no portal login)
 - GET /q/{token} → HTML view
 - POST /q/{token}/respond → STATUS=APPROVED
4. Convert to Sales Order
 - POST /sales-orders/from-quotation → SO DRAFT
5. Send advance request
 - POST /sales-orders/{id}/confirm → STATUS=AWAITING_ADVANCE
 - Email with advance amount + due date
 - MD WhatsApp alert
6. Customer pays advance
 - POST /sales-orders/{id}/payment → ADVANCE_RECEIVED += amount
 - Auto-confirm when ≥ required advance
 - MD WhatsApp alert
7. Start production
 - POST /sales-orders/{id}/start-production
 - Projects spawned per line
 - Work Orders created
 - STATUS=IN_PRODUCTION
8. Production floor
 - PATCH /process/wo/{wo}/stages/{stage} per stage completion
9. QC gate
 - POST /quality/inspections + finalise → blocks WO DONE until passed
10. Ship → Deliver → Close

2.6 Integration Map

Integration	Direction	Used For
Resend / SMTP	Outbound	All transactional emails
WhatsApp Business API	Outbound	MD revenue-event alerts
CallMeBot	Outbound	Free WhatsApp fallback

Integration	Direction	Used For
Google Gemini	Outbound	General ERP chatbot
Biometric devices	Inbound	Attendance from raw scans
Browser Print API	Client	A4 PDF generation

2.7 Stateless Backend

The FastAPI process is stateless — restartable at any time, horizontally scalable behind Nginx. The auto-migration block is idempotent (`IF NOT EXISTS`) so deployments don't need separate schema scripts.

3. Technology Stack

3.1 Backend

Layer	Technology	Version
Language	Python	3.13
Web framework	FastAPI	latest
ASGI server	Uvicorn	latest
ORM	SQLAlchemy	2.x
DB driver	PyMySQL	latest
Validation	Pydantic	v2
Password hashing	bcrypt (<code>passlib</code>)	latest
JWT	<code>python-jose</code>	latest
Email	Resend HTTP API / SMTP	—
WhatsApp	Cloud API + CallMeBot	v22.0
Chat	Google Gemini	v1

3.2 Frontend

Dependency	Version	Purpose
React	^19.2.6	UI

Dependency	Version	Purpose
React DOM	^19.2.6	rendering
React Router DOM	^7.15.0	routing
Axios	^1.16.0	HTTP
Recharts	^3.8.1	charts
Vite	latest	bundler

No external UI library, no Tailwind. The visual layer is custom CSS-in-JS with BVC red. Bundle size ~250 KB gzipped.

3.3 Database

- **MySQL 8.x**, `utf8mb4` , `utf8mb4_unicode_ci` .
- 50+ tables, all with `VENDOR_ID` .
- Indices on every `*_ID` foreign key and commonly filtered status columns.
- Auto-migration on backend startup (idempotent `ALTER TABLE ... IF NOT EXISTS`).

3.4 Third-Party Services

Service	Required	Notes
Resend or SMTP	One required	Email transport
WhatsApp Cloud API / CallMeBot	Optional	MD alerts
Google Gemini	Optional	General chatbot
Biometric device	Optional	Attendance

All optional services degrade gracefully — missing config never blocks a business operation.

4. Installation & Deployment

4.1 Prerequisites

Software	Min. version
Python	3.11+ (3.13 recommended)
Node.js	20 LTS

Software	Min. version
MySQL	8.0+
Git	any

4.2 Local Development

```
# 1. Clone
git clone <repo> bvc24-erp && cd bvc24-erp

# 2. Database
mysql -u root -p <<EOF
CREATE DATABASE bvc24 CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER 'bvc24'@'localhost' IDENTIFIED BY 'CHANGE_ME';
GRANT ALL ON bvc24.* TO 'bvc24'@'localhost';
EOF

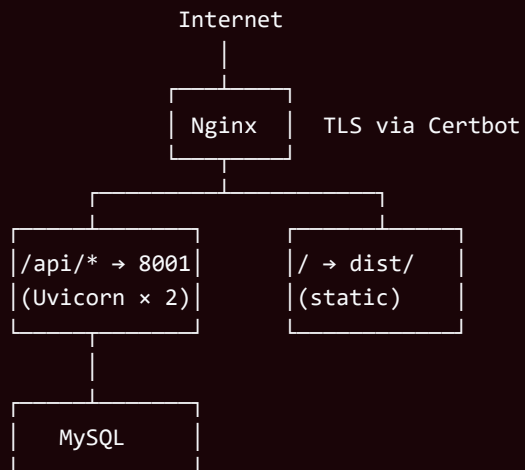
# 3. Backend
cd backend
python -m venv venv && venv/Scripts/activate
pip install -r requirements.txt
# Create .env per Appendix C
uvicorn app.main:app --host 0.0.0.0 --port 8001 --reload

# 4. Seed
# Visit http://127.0.0.1:8001/docs and POST:
#   /seed-org, /seed-admin, /seed-employees, /seed-bvc24,
#   /seed-materials, /seed-project-templates

# 5. Frontend
cd ../frontend
npm install
npm run dev      # → http://localhost:5173
```

4.3 Production Deployment

Single-VM topology (4 vCPU / 8 GB RAM) is sufficient for the current scale.



4.4 Systemd unit (backend)

```
[Unit]
Description=BVC24 Backend
After=network.target mysql.service

[Service]
Type=simple
User=www-data
WorkingDirectory=/opt/bvc24-erp/backend
ExecStart=/opt/bvc24-erp/backend/venv/bin/uvicorn app.main:app \
    --host 127.0.0.1 --port 8001 --workers 2
Restart=always

[Install]
WantedBy=multi-user.target
```

4.5 Nginx (frontend + API proxy)

```
server {
    listen 80;
    server_name erp.bvc24.in;

    root /opt/bvc24-erp/frontend/dist;
    index index.html;
    try_files $uri /index.html;

    location /api/ {
        proxy_pass http://127.0.0.1:8001/;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-Proto $scheme;
    }

    location /static/ {
        proxy_pass http://127.0.0.1:8001/static/;
    }
}
```

4.6 Upgrade Procedure

```
cd /opt/bvc24-erp
git pull origin main
cd backend && venv/bin/pip install -r requirements.txt
cd ../frontend && npm ci && npm run build
sudo systemctl restart bvc24-backend
```

Auto-migration applies any new schema changes idempotently on startup.

4.7 Backups

Daily mysqldump + tarball of `backend/app/static/` . Retain 30 days + 1 monthly archive.

4.8 Required Environment Variables (minimal)

```
MY_SQL=localhost:3306
DB_NAME=bvc24
SECRET_KEY=<random 32+ char>
RESEND_API_KEY=re... # OR SMTP_HOST/PORT/USER/PASSWORD
SMTP_FROM=erp@bvc24.in
APPROVER_EMAIL=md@bvc24.in
FRONTEND_URL=https://erp.bvc24.in
BACKEND_URL=https://erp.bvc24.in/api
```

See Appendix C for the complete env-var reference.

5. Security & Authentication

5.1 Auth Model

JWT-based stateless authentication with two login flows:

Role	Endpoint	Identity
Admin / MD / HR / Production Head	POST /admin-login	EMAIL + password
Floor Employee	POST /employee-login	EMPLOYEE_CODE + password

Tokens signed with `SECRET_KEY` (HS256). Default expiry 7 days. 401 → client clears `localStorage` and redirects to `/login`.

5.2 Password Storage

bcrypt via `passlib`, cost 12. No plaintext stored. `PUT /employees/{id}/reset-password` lets admin reset.

5.3 Role-Based Access Control

Seeded system roles: `SUPER_ADMIN`, `ADMIN`, `HR`, `MANAGER`, `PRODUCTION_HEAD`, `SALES`, `EMPLOYEE`.

Permission codes (`task.assign`, `report.export`, etc.) are defined and stored, with role-level enforcement at routes today. Per-permission decorators are on the roadmap.

5.4 Multi-tenancy

Every customer-facing table includes `VENDOR_ID`. JWT carries `vendor_id`. Every list query filters by it. No shared data between tenants.

5.5 Public Endpoints (No Authentication)

Endpoint	Auth
POST /admin-login, /login, /employee-login	Credentials
GET /q/{token}	Token in URL (quotation public view)
POST /q/{token}/respond	Token in URL
GET /approve-task?token=...	Token in URL
GET /reject-task?token=...	Token in URL

Endpoint	Auth
GET /leave/decide/{token}?action=	Token in URL
POST /biometric/scan	(device, consider device-token in production)

Tokens are `secrets.token_urlsafe(24)` — 32 URL-safe characters. Inert after consumption.

5.6 Transport & Storage

- TLS via Nginx + Certbot.
- bcrypt password hashing.
- All SQL queries parameterised via SQLAlchemy ORM.
- React escapes output by default.
- CORS configured in `main.py`; restrict `allow_origins` in production.

5.7 Audit Trail

Every business-critical entity has an `*_activity` companion table with `EVENT_TYPE`, `EVENT_DETAIL`, `ACTOR_TYPE` (`SYSTEM` / `SALES` / `CUSTOMER` / `SUPPLIER` / `WAREHOUSE` / `ADMIN`), `ACTOR_NAME`, `CREATED_AT`.

5.8 Production Hardening Checklist

- [] Set `SECRET_KEY` to a 32+ char random value.
- [] Restrict CORS `allow_origins` to production frontend host.
- [] Gate seed endpoints behind a production check.
- [] Force HTTPS at Nginx.
- [] Configure DB backups offsite.
- [] Configure log shipping (journal → ELK / CloudWatch).
- [] Add rate limiting on login endpoints.

6. Database Schema

A complete data dictionary follows. Generic columns (`ID`, `CREATED_AT`, `UPDATED_AT`) are omitted unless they carry notable defaults.

6.1 Organization & Multi-tenancy

Model	Table	Notes
Vendor	vendor	Tenant; VENDOR_NAME
RootUser	root_user	Tenant admin; EMAIL , PASSWORD
Department	department	CODE , NAME , HEAD_EMPLOYEE_ID
Designation	designation	TITLE , DEPARTMENT_ID , BASE_SALARY
Role	role	ROLE_NAME , IS_SYSTEM
Permission	permission	CODE , NAME
RolePermission	role_permission	M2M role↔permission

6.2 People

Employee (employee) — primary fields: EMPLOYEE_CODE (unique), NAME , EMAIL , PHONE , PASSWORD (bcrypt), DEPARTMENT_ID , DESIGNATION_ID , ROLE_ID , REPORTING_MANAGER_ID (self-FK), JOINING_DATE , SALARY , SHIFT_START / END , STATUS (ACTIVE/SUSPENDED/RESIGNED/TERMINATED), FINGERPRINT_ID , PROFILE_SUBMITTED , SKILLS . Profile: ADDRESS , CITY , STATE , PINCODE , DOB , GENDER , FATHER_NAME , MOTHER_NAME , MARITAL_STATUS , OCCUPATION , QUALIFICATION , YEAR_OF_PASSING , EXPERIENCE_YEARS , EXPERIENCE_DETAILS , PAST_PROJECTS , EMPLOYMENT_TYPE , PHOTO_URL . VENDOR_ID .

Customer (customer): CUSTOMER_CODE , CUSTOMER_NAME , CONTACT_PERSON , PHONE , EMAIL , GST_NUMBER , INDUSTRY , STATUS (LEAD/PROSPECT/ACTIVE/INACTIVE), LEAD_STATUS (NEW/QUALIFIED/PROPOSAL/NEGOTIATION/CLOSED_WON/CLOSED_LOST), LEAD_SOURCE , LEAD_PRIORITY , ASSIGNED_SALES_ID , qualification fields (NUMBER_OF_BRANCHES , EXPECTED_MONTHLY_ORDERS , EXISTING_MACHINE_USAGE , CURRENT_VENDOR_NAME), BILLING_ADDRESS , SHIPPING_ADDRESS , WHATSAPP_NUMBER .

Supplier (supplier): SUPPLIER_CODE , COMPANY_NAME , CONTACT_PERSON , PHONE , EMAIL , GST_NUMBER , CATEGORY , PAYMENT_TERMS , STATUS (ACTIVE/INACTIVE/BLACKLISTED).

6.3 Sales

Quotation (quotation): QUOTATION_NUMBER (unique), CUSTOMER_ID , QUOTATION_DATE , VALIDITY_DAYS , EXPIRY_DATE , STATUS (DRAFT/SENT/APPROVED/REJECTED/CONVERTED/EXPIRED), totals (SUBTOTAL , DISCOUNT_PERCENT , TAX_PERCENT , GRAND_TOTAL), PREPARED_BY , PUBLIC_TOKEN , EMAIL_SENT_AT , VIEWED_AT , VIEW_COUNT .

QuotationLine (quotation_line): QUOTATION_ID , PRODUCT_MODEL_ID , REQUIREMENT_ID , DESCRIPTION , HSN_CODE , QUANTITY , UNIT , UNIT_PRICE , DISCOUNT_PERCENT , LINE_TOTAL .

QuotationActivity (quotation_activity): audit timeline.

SalesOrder (`sales_order`): `SO_NUMBER` (unique), `CUSTOMER_ID` , `QUOTATION_ID` , `SO_DATE` , `EXPECTED_DELIVERY_DATE` , `ADVANCE_DUE_DATE` (Phase 5), `STATUS` (DRAFT / **AWAITING_ADVANCE** / **CONFIRMED** / **IN_PRODUCTION** / **SHIPPED** / **DELIVERED** / **CLOSED** / **CANCELLED**), totals, `ADVANCE_PERCENT` (50)/ `DISPATCH_PERCENT` (40)/ `INSTALLATION_PERCENT` (10), `ADVANCE_RECEIVED` / `DISPATCH_RECEIVED` / `INSTALLATION_RECEIVED` , `PREPARED_BY` , `CONFIRMED_AT` , `PRODUCTION_STARTED_AT` , `SHIPPED_AT` , `DELIVERED_AT` , `CLOSED_AT` , `CANCELLED_AT` , `CANCEL_REASON` .

SalesOrderLine (`sales_order_line`): `SO_ID` , `PRODUCT_MODEL_ID` , `QUOTATION_LINE_ID` , `SPAWNED_PROJECT_ID` (set when production starts), pricing, `LINE_TOTAL` .

SalesOrderActivity (`sales_order_activity`): audit timeline.

6.4 Purchase

PurchaseOrder (`purchase_order`): `PO_NUMBER` , `SUPPLIER_ID` , `PO_DATE` , `EXPECTED_DELIVERY_DATE` , `STATUS` (DRAFT/SENT/CONFIRMED/PARTIAL_RECEIVED/RECEIVED/CANCELLED), totals, `DELIVERY_ADDRESS` , `LINKED_PROJECT_ID` , `PREPARED_BY` .

PurchaseOrderLine (`purchase_order_line`): `PO_ID` , `MATERIAL_ID` , `BOM_ITEM_ID` , `QUANTITY` , `QUANTITY_RECEIVED` , pricing.

GoodsReceiptNote (`goods_receipt_note`): `GRN_NUMBER` , `PO_ID` , `RECEIVED_DATE` , `RECEIVED_BY` , `STATUS` (DRAFT/FINAL), `INVOICE_NUMBER` .

GoodsReceiptLine (`goods_receipt_line`): `GRN_ID` , `PO_LINE_ID` , `QUANTITY_RECEIVED` , `QUANTITY_REJECTED` , `REJECTION_REASON` .

PurchaseOrderActivity (`purchase_order_activity`).

6.5 Inventory

MaterialCatalog (`material_catalog`): master list — `MATERIAL_NAME` unique.

MaterialDepartment (`material_department`): M2M material↔department.

Inventory (`inventory`): per-vendor stock — `MATERIAL_ID` , `MATERIAL_NAME` , `QUANTITY` , `UNIT_PRICE` .

6.6 Production

ProductModel (`product_model`): `MODEL_NAME` , `MODEL_CODE` , `CATEGORY` , `ESTIMATED_BUILD_DAYS` , `STATUS` .

BOMItem (`bom_item`): `PRODUCT_MODEL_ID` , `MATERIAL_ID` , `MATERIAL_NAME` , `QUANTITY` , `UNIT` , `ITEM_TYPE` (PURCHASE/PROCESS), `PREFERRED_SUPPLIER_ID` , `PROCESS_STAGE_ID` , `ITEM_NO` , `IMAGE_URL` .

ProcessStage (`process_stage`): `PRODUCT_MODEL_ID` , `SEQUENCE` , `STAGE_NAME` , `STAGE_TYPE` (DESIGN / MECHANICAL / ELECTRICAL / WIRING / FABRICATION / ASSEMBLY / TESTING / QC / PACKAGING / OTHER), `ESTIMATED_HOURS` , `IS_ACTIVE` .

WorkOrder (*work_order*): *WO_NUMBER* , *PRODUCT_MODEL_ID* , *PROJECT_ID* , *QUANTITY* , *STATUS* (PLANNED/IN_PROGRESS/ON_HOLD/DONE/CANCELLED), planned + actual dates.

WorkOrderStageProgress (*wo_stage_progress*): *WORK_ORDER_ID* , *STAGE_ID* , *STATUS* (PENDING/IN_PROGRESS/DONE/FAILED/SKIPPED), *ASSIGNED_TO_ID* , *STARTED_AT* , *COMPLETED_AT* , *NOTES* .

Machine (*machine*): *MACHINE_NAME* , *MACHINE_TYPE* , *STATUS* , *LOCATION* , *PRODUCT_MODEL_ID* , *WORK_ORDER_ID* , *UNIT_NUMBER* , *SERIAL_NO* .

MachineLog (*machine_log*): *MACHINE_ID* , *STATUS* , *NOTE* , *TIMESTAMP* .

6.7 Quality

QCChecklistItem (*qc_checklist_item*): *PRODUCT_MODEL_ID* , *SEQUENCE* , *CHECK_POINT* , *DESCRIPTION* , *SEVERITY* (CRITICAL/MAJOR/MINOR), *IS_ACTIVE* .

QCInspection (*qc_inspection*): *WORK_ORDER_ID* , *PRODUCT_MODEL_ID* , *INSPECTOR_ID* , *INSPECTION_DATE* , *STATUS* , counts.

QCInspectionResult (*qc_inspection_result*): *INSPECTION_ID* , *CHECKLIST_ITEM_ID* , *CHECK_POINT* , *RESULT* (PASS/FAIL/NEEDS_REWORK/NA), *NOTES* .

NCR (*ncr*): auto-created on failure — *NCR_NUMBER* , *INSPECTION_ID* , *WORK_ORDER_ID* , *PRODUCT_MODEL_ID* , *CHECK_POINT* , *SEVERITY* , *DESCRIPTION* , *ROOT_CAUSE* , *CORRECTIVE_ACTION* , *STATUS* (OPEN/IN_PROGRESS/CLOSED), assignees, timestamps.

6.8 Projects & Tasks

ProjectCategory (*project_category*): *SECTION* , *NAME* .

SubProjectTemplate (*sub_project_template*): *CATEGORY_ID* , *NAME* , *ESTIMATED_TOTAL_DAYS* .

Project (*project*): *PROJECT_NAME* , *DESCRIPTION* , *STATUS* , *SUB_PROJECT_TEMPLATE_ID* , *DEPARTMENT_ID* , *CUSTOMER_ID* , *PRODUCT_MODEL_ID* , *QUANTITY* , *TARGET_DATE* , *PRIORITY* , *SKILLS_REQUIRED* .

Task (*task*): *TASK_NAME* , *STATUS* , *PRIORITY* , *PROJECT_ID* , *ASSIGNED_TO* , *START_TIME* , *END_TIME* .

TaskAssignment (*task_assignment*): formal allocation with *APPROVAL_STATUS* (PENDING_APPROVAL/APPROVED/REJECTED/EXPIRED), *APPROVAL_TOKEN* , *ASSIGNED_BY_ID* .

DailyAllocation (*daily_allocation*): AI allocator output per employee per day.

6.9 Attendance & Leave

BiometricEvent (*biometric_event*): raw device scan log.

Attendance (*attendance*): daily record — *EMPLOYEE_ID* , *DATE* , *CHECK_IN* , *CHECK_OUT* , *STATUS* , *WORKED_HOURS* , *OVERTIME_HOURS* .

LeaveRequest (*leave_request*): *EMPLOYEE_ID* , *LEAVE_TYPE* (CASUAL/SICK/EARNED/UNPAID/LOP), dates, *DAYS* , *REASON* , *STATUS* , *APPROVAL_TOKEN* .

LeaveBalance (`leave_balance`): annual quota — Casual 12, Sick 12, Earned 15.

6.10 Payroll & Performance

PayrollRun (`payroll_run`): header per (vendor, year, month) — `STATUS` (DRAFT/FINALIZED/PAID).

PayrollSlip (`payroll_slip`): per-employee breakdown with attendance counts, task bonus, OT, penalties, net pay.

PerformanceScore (`performance_score`): monthly STAR rating with weighted formula (25 % attendance + 30 % tasks + 25 % productivity + 20 % consistency), recommendations.

6.11 System

Setting, Notification.

Part II — Functional Modules

Module 01 — Organization Management

Defines Vendor, Department, Designation, Role, Permission, RolePermission.

Workflows:

- Initial setup: `POST /create-vendor` → `POST /seed-org` (MANUFACTURING preset) → `POST /seed-admin`.
- Custom departments / roles / permissions via standard CRUD endpoints.

System roles seeded: `SUPER_ADMIN`, `ADMIN`, `HR`, `MANAGER`, `PRODUCTION_HEAD`, `SALES`, `EMPLOYEE` — `IS_SYSTEM=1` prevents deletion.

Permissions follow `<entity>.<action>` codes (e.g. `task.assign`). Definitions are seeded; enforcement is currently role-level.

Multi-tenancy: every customer-facing table carries `VENDOR_ID`; JWT carries `vendor_id`; queries filter accordingly.

Module 02 — Human Resources

Five sub-modules sharing the `employee` table.

2.1 Employee Master

- Directory at `/employees` with photos, skill tags.
- Self-registration flow: admin creates employee → employee logs in → sees `EmployeeProfileForm` → submits → `PROFILE_SUBMITTED=1` locks the form.
- ResumeModal shows full profile with sticky header.

2.2 Attendance

- `/biometric` kiosk for fingerprint check-in.
- Device pushes `/biometric/scan` → BiometricEvent → resolved to Attendance via `FINGERPRINT_ID`.
- Status (PRESENT/LATE/ABSENT/HALF_DAY) computed from check-in vs. shift start.

2.3 Leave

- Token-based email approval — single click from approver's email.
- LeaveBalance tracks Casual (12), Sick (12), Earned (15) annual quota.
- Casual/Sick auto-convert to LOP when exhausted; Earned rejected.

2.4 Payroll

- Auto-computed from attendance + completed tasks + OT.
- Formula: $\text{EARNED_BASIC} + \text{TASK_BONUS} + \text{OT_PAY} - \text{DEDUCTIONS} = \text{NET_PAY}$.
- $\text{TASK_BONUS} = \text{completed tasks} \times 100$ (default).
- Runs progress DRAFT → FINALIZED → PAID.

2.5 STAR Performance

- Monthly star rating per employee.
- Formula: $0.25 \times \text{ATT} + 0.30 \times \text{TASK} + 0.25 \times \text{PROD} + 0.20 \times \text{CONS}$.
- ≥ 4.5 → recommended for promotion; ≥ 4.0 → recommended for increment.
- MD approves recommendations via MDReview screen.

Module 03 — CRM

Customer + Contact + Requirement entities. Lead pipeline:

NEW → QUALIFIED → PROPOSAL → NEGOTIATION → CLOSED_WON / CLOSED_LOST

WhatsApp MD alerts fire on `POST /create-customer` and `POST /customers/enquiry` with rich context (customer, industry, sales rep, expected orders).

Convert path: `POST /customers/{cid}/requirements/{rid}/to-project` skips quotation and spawns a project directly.

Qualification fields (`BUSINESS_TYPE` , `NUMBER_OF_BRANCHES` , `EXPECTED_MONTHLY_ORDERS` , `LEAD_PRIORITY`) help prioritise leads.

Module 04 — Quotations

State machine: `DRAFT → SENT → APPROVED/REJECTED → CONVERTED` (or `EXPIRED`).

Public link — `GET /q/{token}` serves a no-auth customer view; `POST /q/{token}/respond` records approval/rejection. Telemetry: `VIEWED_AT` , `VIEW_COUNT` .

Auto-pricing: `POST /quotations/from-requirements` builds a quote from BOM with a configurable margin.

Email: HTML template with BVC red header, line items, totals, validity warning, CTA to public link.
Transport: Resend → SMTP fallback.

Print: `/quotation-print/:id` renders A4 layout for browser PDF export.

Activity timeline records `CREATED`, `LINE_*`, `SENT`, `EMAIL_SENT/FAILED`, `VIEWED`, `APPROVED`, `REJECTED`, `CONVERTED`.

Module 05 — Sales Orders (Phase 5)

The most business-critical module.

Payment-gated state machine

```
DRAFT —/confirm→▶ AWAITING_ADVANCE —advance fully paid→▶ CONFIRMED
                                     |
                                     ▼
                                     (auto on /payment)
CONFIRMED —/start-production→▶ IN_PRODUCTION —/ship→▶ SHIPPED
          —/deliver→▶ DELIVERED —/close→▶ CLOSED
(any non-CLOSED) —/cancel→▶ CANCELLED
```

Three payment milestones (defaults)

- Advance 50 % — before production starts

- On Dispatch 40 % — at shipping
- On Installation 10 % — at site delivery

Workflow

1. `POST /sales-orders/{id}/confirm` — sets `AWAITING_ADVANCE`, ensures `ADVANCE_DUE_DATE` (defaults `SO_DATE + 7 days`), emails customer with **highlighted advance amount + due date box**.
2. `POST /sales-orders/{id}/payment ( MILESTONE=ADVANCE )` — increments `ADVANCE_RECEIVED`. **Auto-flips to CONFIRMED** when $\geq$ required advance.
3. `POST /sales-orders/{id}/start-production` — invokes `project_from_product_service.create_project_from_product()` for each line, spawning Project + Tasks + Work Order. `SPAWNED_PROJECT_ID` recorded on the SO line.
4. `/ship` → SHIPPED, `/deliver` → DELIVERED, `/close` → CLOSED.

MD WhatsApp alerts

- `/confirm` → " Sales Order — Awaiting Advance" (SO #, customer, total, advance amount + due).
- `/payment` auto-confirm → " Sales Order CONFIRMED — Advance Received".
- `from-quotation` → " Quotation CONVERTED to Sales Order".

UI

- Status pill, action button changes by state.
- Amber `AWAITING_ADVANCE` banner showing advance ₹ + due date.
- Payment progress card with three milestone tiles.

Computed fields on `GET /sales-orders/{id}`

`ADVANCE_AMOUNT`, `DISPATCH_AMOUNT`, `INSTALLATION_AMOUNT`, `PAYMENT_RECEIVED_TOTAL`, `PAYMENT_PENDING`, `PAYMENT_PROGRESS_PCT`.

Module 06 — Purchase Orders

State machine: `DRAFT` → `SENT` → `CONFIRMED` → `PARTIAL_RECEIVED` → `RECEIVED` (or `CANCELLED`).

GRN workflow: a single PO may have multiple GRNs (partial deliveries). Each GRN tracks `QUANTITY_RECEIVED` + `QUANTITY_REJECTED` per line. GRN starts as `DRAFT`; **only FINAL GRNs push to inventory**.

Auto-PO from project: `POST /purchase-orders/auto-from-project` walks the BOM, groups by `PREFERRED_SUPPLIER_ID`, creates one PO per supplier. A 30-item BOM across 5 suppliers becomes 5 POs in one click.

Rejection notice: `POST /purchase-orders/grn/{id}/resend-rejection-notice` emails the supplier with itemised rejections + reasons.

Activity timeline records the full chain — CREATED, LINE_*, SENT, CONFIRMED, GRN_RECORDED, GRN_FINALIZED, PARTIAL_RECEIVED, RECEIVED, REJECTION_NOTICE_SENT, CANCELLED.

Module 07 — Suppliers

Master record of all external vendors. `STATUS = ACTIVE / INACTIVE / BLACKLISTED`. PO creation validates supplier is `ACTIVE`.

Categories: Sheet Metal, Electronics / PCBs, Refrigeration, Vending Mechanisms, Payment Systems, Display, Packaging, Logistics, Misc.

360° view (`/connect/supplier/{id}/360`) aggregates total POs, on-time delivery rate, GRN rejection rate, average order value, last 10 POs.

Module 08 — Inventory

KPI tiles: Total Materials, Total Stock Value, Low Stock, Out of Stock.

Status buckets: `OUT_OF_STOCK` (qty ≤ 0), `LOW_STOCK` (qty < reorder), `OK`.

Category auto-detection from material name maps to emoji badges (📏 Sheet Metal, 🧊 Refrigeration, ⚡ Electrical, etc.).

Stock adjustment (`POST /inventory/{id}/adjust`) — TYPE = RECEIPT / ISSUE / COUNT. Each adjustment logs a movement.

Inventory ← **GRN pipeline:** only GRN finalisation pushes stock. Drafts don't affect inventory.

Material → **Department scoping:** `/materials/for-me` returns only materials accessible to the current employee's department.

Module 09 — Projects & Tasks

Projects are execution containers between sale and delivered machine.

Spawning paths:

1. Primary — `POST /sales-orders/{id}/start-production` via `project_from_product_service`.
2. CRM shortcut — `POST /customers/{cid}/requirements/{rid}/to-project`.
3. Manual — `POST /create-project`.

Tasks are the unit of daily work. Lifecycle: `PENDING → IN_PROGRESS → COMPLETED` (or `ON_HOLD`).

TaskAssignment adds formal approval workflow:

- `APPROVAL_TOKEN` for one-click email approval / rejection.
- Approver gets `/approve-task?token=...` / `/reject-task?token=...` links.
- Employee then accepts / rejects from their dashboard.

Workload service picks the least-loaded employee matching required skills for auto-assignment.

DailyAllocation is the AI allocator's output — which employee gets which task on which day.

Carryover banner — `pending-from-yesterday` surfaces incomplete tasks the next morning.

Module 10 — Production & BOM

Product Models

The vending machine catalog: `BVC24 Snack & Beverage Combo (BVC-SBC-01)` , `BVC24 Medicine Dispenser (BVC-MED-01)` , etc.

BOM (Bill of Materials)

Each line is classified as `ITEM_TYPE` :

- **PURCHASE** — sourced from supplier (feeds auto-PO flow).
- **PROCESS** — made in-house at a specific process stage.

Each line carries `IMAGE_URL` (assembly reference photo).

`POST /production/models/{id}/seed-default-bom` creates a 10-stage flow: Design Review → Mechanical Design → Electrical Design → Sheet Metal Fabrication → Electrical Wiring → Component Assembly → Software Flashing → Bench Testing → Pre-Dispatch QC → Packaging & Dispatch (86h total).

Process Stages

Colour-coded by `STAGE_TYPE` :

Type	Colour
DESIGN	Purple
MECHANICAL	Blue
ELECTRICAL	Cyan
FABRICATION	Amber
ASSEMBLY	Emerald

Type	Colour
TESTING	Pink
QC	Red
PACKAGING	Slate

Work Orders

State machine: `PLANNED` → `IN_PROGRESS` → `DONE` (or `ON_HOLD` / `CANCELLED`). On create, `POST /process/spawn-for-wo/{wo_id}` spawns one `WorkOrderStageProgress` row per active stage.

Gantt Timeline (`GET /process/wo/{wo_id}/gantt`)

The marquee feature. Returns a 30-day calendar timeline with:

- `timeline` — array of 30 day-cells with `day_number`, `date`, `weekday`, `is_sunday`.
- `stages` — each carries `day_number`, `planned_start_date`, `planned_end_date`, `days_allocated: 1`, `assignee_name`, `assignee_code`.

Scheduling rule: Day 1 = `WO.PLANNED_START_DATE`. One stage = one working day. Sundays skipped.

Auto-assignment policy: the endpoint **does not auto-assign**. It **does clear stale assignments to sales-role employees** (name/role/dept/designation/skills matching `sales` / `marketing` / `ragul`).

Gantt UI

- Sticky header — WO number, product, units always visible.
- 4 summary tiles — Progress %, Planned days, Actual So Far, Failed Stages.
- Day 1–30 calendar header with date and weekday under each day number; Sundays in red.
- Per-row card shows stage name, type chip, hours, `1 day`, assignee chip, status pill, action buttons (▶ ✓ X).
- Drawer width `98vw` so all 30 day cells fit without horizontal scroll.

Module 11 — Quality Management

Three-tier model: **template** (`QCChecklistItem`) → **instance** (`QCInspection`) → **result** (`QCInspectionResult`).

Workflow:

1. `POST /quality/inspections` creates the inspection + spawns one result row per checklist item.
2. Per-item `PATCH /quality/results/{id}` records `PASS` / `FAIL` / `NEEDS_REWORK` / `NA`.

3. Failures auto-create NCRs.

4. `POST /quality/inspections/{id}/finalise` locks the inspection. Status computed: - All PASS → PASS - Any FAIL → FAIL - Only NEEDS_REWORK → REWORK

Severity policy:

Severity	Failure consequence
CRITICAL	Auto-NCR, blocks WO DONE , requires re-inspection
MAJOR	Auto-NCR, blocks WO DONE
MINOR	Auto-NCR, does NOT block WO DONE

NCR resolution: OPEN → IN_PROGRESS → CLOSED . Closure requires ROOT_CAUSE and CORRECTIVE_ACTION .

WO → DONE gate: all stages DONE + inspection passed + open NCRs closed.

Module 12 — Machines

Registry of manufactured units. **Auto-registered** when a Work Order completes — one Machine row per unit produced.

Status flow: IDLE → ACTIVE → MAINTENANCE → ACTIVE (or decommissioned).

MachineLog is the append-only history of status changes. Used as service history until Phase 7 formalises it into structured ServiceVisit records.

`POST /machines/sync` bulk-registers machines for historical WOs that pre-date the auto-registration logic (idempotent).

Module 13 — Notifications

Four channels, each with graceful degradation.

Email

`email_service.py` tries Resend → SMTP fallback. Email types: quotation send, SO advance request, PO send, GRN rejection notice, leave / task approval links.

Dev override: `EMAIL_TESTING_OVERRIDE_TO` redirects all email to one address.

WhatsApp

`whatsapp_service.py` tries CallMeBot (free) → WhatsApp Business Cloud API (paid). `notify_md_safe()` is fire-and-forget — never blocks parent op.

MD alert triggers: new customer, new enquiry, quotation → SO conversion, SO awaiting advance, SO auto-confirmed.

In-app

`notification` table. NotificationBell component polls `/notifications/unread-count` every 10 s.

Voice

Opt-in browser TTS (no external service). Speaks critical alerts (out-of-stock, failed QC, etc.). Employee dashboard variant reads today's task list on login.

Module 14 — Chatbot & HR Assistant

Two bots intentionally separated.

General ERP Chatbot

`gemini_service.py` — Google Gemini-backed. Scoped via system prompt to BVC24 questions only. Chat history client-side (not persisted). Falls back to friendly message when `GEMINI_API_KEY` not set.

HR Assistant

`hr_assistant.py` — stateless rule-based bot for leave application. Walks employee through type / dates / reason via dialogue. Validates against actual `LeaveBalance` server-side. No LLM, no hallucination risk.

Both bots use a shared floating widget with BVC red gradient for user bubbles.

Module 15 — Dashboard & Analytics

`DashboardHome` (/) is the executive cockpit:

- 4-grid stat cards (Employees / Projects / Tasks / Inventory).
- Recharts donut + bar visualisations.
- InventorySummaryCard with low-stock items.
- Pending acceptance + today's allocations + recent activity.

Refreshes every 10 s. Opt-in voice alerts.

`GET /analytics/dashboard-stats` returns all KPIs in one round trip.

`GET /analytics/chart-data?range=7d|30d|month` returns time-series.

Reports: `/reports/report/{module}.pdf|.xlsx` for sales / production / inventory / attendance / payroll / performance / quality.

MDReview (`/md-review`) is the MD-specific deeper KPI view (revenue trend, top performers, lead pipeline).

Module 16 — Print & Public Links

Print views (4)

- `/quotation-print/:id`
- `/so-print/:id`
- `/po-print/:id`
- `/grn-print/:id`

Each renders an A4-styled HTML page. Users press Ctrl+P → Save as PDF. No PDF library required.

Public quotation link

`/q/{PUBLIC_TOKEN}` — customer-facing no-auth view + Approve/Reject endpoint. Token is `secrets.token_urlsafe(24)` — unguessable, per-quotation. Telemetry tracked.

Other token-gated routes

- `/approve-task?token=...` , `/reject-task?token=...`
- `/leave/decide/{token}?action=approve|reject`
- `/biometric` (kiosk)
- `/apply-leave` (deep link)

All become inert after consumption.

Part III — API Reference

A complete inventory follows. Endpoints are grouped by router. All endpoints require `Authorization: Bearer <JWT>` unless marked **(no auth)**.

Authentication

Method	Path	Purpose
POST	/admin-login , /login	Admin login
POST	/employee-login	Employee login
POST	/employee-logout	Logout
GET	/me	Verify token

Organization

Method	Path	Purpose
GET / POST / PUT / DELETE	/departments	Department CRUD
GET / POST / PUT / DELETE	/designations	Designation CRUD
GET / POST / DELETE	/roles	Role CRUD
GET	/permissions	List permissions
PUT	/roles/{id}/permissions	Assign permissions
POST	/seed-org	Apply MANUFACTURING preset
GET	/org-presets	List presets
POST	/create-vendor , GET /vendors	Vendor CRUD

HR — Employees

Method	Path	Purpose
POST	/create-employee	Create
GET	/employees , /employees/{id} , /employees/by-code/{code}	Lookup
POST	/employees/by-code/{code}/submit-profile	Self-registration
PUT	/update-employee/{id}	Update
PUT	/employees/{id}/reset-password	Reset password
POST	/employees/{id}/upload-photo	Photo
DELETE	/delete-employee/{id}	Remove

HR — Attendance

Method	Path	Purpose
POST	/attendance/check-in , /check-out , /mark-absent	Manual
GET	/attendance , /attendance/today , /attendance/live-board	List
DELETE	/attendance/{id}	Remove
POST	/biometric/enroll , /biometric/scan	Biometric
GET	/biometric/events	Raw log

HR — Leave

Method	Path	Purpose
POST	/leave/apply	Apply
GET	/leave/decide/{token}?action= (no auth)	Email link decision
PATCH	/leave/{id}/approve , /reject , /cancel	Workflow
GET	/leave/pending , /all , /my-requests , /balance/{id} , /dashboard	Queries

HR — Payroll

Method	Path	Purpose
POST	/payroll/generate	Create run
GET	/payroll/runs , /runs/{id} , /runs/{id}/slip/{eid}	Query
PATCH	/runs/{id}/finalize , /mark-paid	Workflow
DELETE	/runs/{id}	Delete unpaid

HR — Performance

Method	Path	Purpose
POST	/performance/stars/compute	Recalculate STAR
GET	/performance/stars , /top , /employee/{id}/history	Query

Method	Path	Purpose
PATCH	/performance/stars/{id}/action	Approve/deny

CRM

Method	Path	Purpose
POST	/create-customer	Create + WhatsApp alert
GET	/customers , /customers/{id}	Lookup
PATCH	/customers/{id}	Update
DELETE	/delete-customer/{id}	Remove
PATCH	/customers/{id}/lead-status	Pipeline
POST	/customers/enquiry	Enquiry + WhatsApp alert
POST / GET / DELETE	/customers/{id}/contacts[/]{cid}]	Contacts
POST / GET / PATCH / DELETE	/customers/{id}/requirements[/]{rid}]	Requirements
POST	/customers/{cid}/requirements/{rid}/to-project	Convert
GET	/connect/customer/{id}/360	360° view

Quotations

Method	Path	Purpose
POST / GET / PATCH / DELETE	/quotations[/]{id}]	CRUD
POST / PATCH / DELETE	/quotations/{id}/lines[/]{lid}]	Lines
POST	/quotations/{id}/send , /resend-email , /approve , /reject	Workflow
POST	/quotations/from-requirements	Auto-build
GET	/quotations/auto-price	Suggest

Method	Path	Purpose
GET / POST	/q/{token} / /q/{token}/respond (no auth)	Public link
GET / DELETE	/quotations/{id}/activity[/aid]	Timeline

Sales Orders

Method	Path	Purpose
POST / GET / PATCH / DELETE	/sales-orders[/id]	CRUD
POST / PATCH / DELETE	/sales-orders/{id}/lines[/lid]	Lines
POST	/sales-orders/{id}/confirm	Send advance request
POST	/sales-orders/{id}/payment	Record milestone, auto-confirm
POST	/sales-orders/{id}/start-production	Spawn projects
POST	/sales-orders/{id}/ship , /deliver , /close , /cancel	Workflow
POST	/sales-orders/from-quotation	Convert quote
GET / DELETE	/sales-orders/{id}/activity[/aid]	Timeline

Purchase Orders

Method	Path	Purpose
POST / GET / PATCH / DELETE	/purchase-orders[/id]	CRUD
POST / PATCH / DELETE	/purchase-orders/{id}/lines[/lid]	Lines
POST	/purchase-orders/{id}/send , /resend-email , /confirm , /cancel	Workflow
POST	/purchase-orders/{id}/grn	Record GRN
GET	/purchase-orders/{id}/grn , /grn/{gid}	Query
POST	/purchase-orders/grn/{gid}/finalize	Push to inventory

Method	Path	Purpose
POST	/purchase-orders/grn/{gid}/resend-rejection-notice	Notify supplier
DELETE	/purchase-orders/grn/{gid}	Delete draft
POST	/purchase-orders/auto-from-project	Generate from BOM
GET / DELETE	/purchase-orders/{id}/activity[/]{aid}]	Timeline

Suppliers

Method	Path	Purpose
POST / GET / PATCH / DELETE	/suppliers[/]{id}]	CRUD
GET	/suppliers/categories	List categories
GET	/connect/supplier/{id}/360	360° view

Inventory

Method	Path	Purpose
GET	/materials-catalog , /materials/for-me , /materials	List
POST	/create-material , /seed-materials	Create
PUT	/materials-catalog/{id}/departments , /update-stock/{id}	Update
DELETE	/delete-material/{id}	Remove
GET	/inventory/full , /inventory/{id}/movements	Query
POST	/inventory/{id}/adjust	Stock movement

Projects & Tasks

Method	Path	Purpose
POST	/create-project , /projects/from-product , /projects/{id}/backfill-tasks , /projects/auto-assign-missing	Create / spawn

Method	Path	Purpose
GET / PATCH / DELETE	/projects[/]{id}]	CRUD
POST	/create-task	Task
GET	/tasks , /project/{id}/tasks	List
PUT	/start-task/{id} , /complete-task/{id} , /hold-task/{id}	Status
POST / PUT / PATCH / DELETE	/task-assignment[/]{id}]	Assignment
GET / PATCH	/employee/{ref}/today-task , /tasks , /pending-acceptance , /pending-from-yesterday	Employee views
GET	/approve-task?token=... , /reject-task?token=... (no auth)	Email link
POST	/task-proposals/cleanup-expired	Cron
GET	/connect/project/{id}/360	360° view

Production & Process

Method	Path	Purpose
POST / GET / PATCH / DELETE	/production/models[/]{id}]	Models
POST	/production/models/{id}/seed-default-bom	Seed 10-stage flow
POST / GET / PATCH / DELETE	/production/models/{id}/bom , /production/bom/{id}	BOM
POST	/production/bom/{id}/upload-image	Image
PATCH	/bom-items/{id}/classify	Set ITEM_TYPE
POST / GET / PATCH / DELETE	/stages[/]{id}]	Process stages
POST / GET / PATCH / DELETE	/production/work-orders[/]{id}]	Work Orders
POST	/process/spawn-for-wo/{id}	Spawn stage progress
GET / PATCH	/process/wo/{id}/stages[/]{sid}]	Stage progress
GET	/process/wo/{id}/ganttt	Gantt timeline

Method	Path	Purpose
GET	/production/dashboard	KPIs

Quality

Method	Path	Purpose
POST / GET / PATCH / DELETE	/quality/checklist-items , /checklist/{model_id}	Checklists
POST / GET / PATCH	/quality/inspections[/]{id}] , /results/{id}	Inspections
POST	/quality/inspections/{id}/finalise	Lock + NCRs
GET / PATCH	/quality/ncrs[/]{id}]	NCRs
GET	/quality/dashboard	KPIs

Machines

Method	Path	Purpose
POST / GET / PUT / DELETE	/machines[/]{id}]	CRUD
POST	/machines/sync	Bulk register
GET	/machines/machine-logs/{id}	History

Notifications

Method	Path	Purpose
POST / GET / DELETE	/notifications[/]{id}]	CRUD
GET	/notifications/unread-count	Badge
PUT	/notifications/{id}/read , /mark-all-read	Mark read
POST	/notifications/generate	System auto

WhatsApp

Method	Path	Purpose
GET	/whatsapp/diagnose	Status
POST	/whatsapp/test	Send test

Chat / HR Assistant

Method	Path	Purpose
POST	/chat , /chat/stream	Gemini chat
GET	/chat/health , /chat/suggestions	Status
POST	/hr-bot/message	HR conversation
GET	/hr-bot/policy , /hr-bot/diagnose	Status

Connect (360° views)

Method	Path	Purpose
GET	/connect/employee/{id}/360	Employee 360
GET	/connect/project/{id}/360	Project 360
GET	/connect/customer/{id}/360	Customer 360
GET	/connect/work-order/{id}/360	WO 360
GET	/connect/supplier/{id}/360	Supplier 360
GET	/connect/workflow/snapshot	Global snapshot

Analytics / Reports / Settings

Method	Path	Purpose
GET	/analytics/dashboard-stats , /chart-data	Dashboard
GET	/reports/report/{module}.pdf , .xlsx	Reports
GET / PUT	/settings , /settings/email-alerts	Settings

Method	Path	Purpose
POST	/settings/test-email	Test

Part IV — Appendix

A. Glossary

Term	Meaning
AMC	Annual Maintenance Contract
BOM	Bill of Materials
BVC	Bharath Vending Corporation
BVC24	The product / system name
CallMeBot	Free WhatsApp bridge
CRM	Customer Relationship Management
ERP	Enterprise Resource Planning
GRN	Goods Receipt Note
GST	Indian Goods & Services Tax
HSN	Harmonized System of Nomenclature
JWT	JSON Web Token
LOP	Loss of Pay
MD	Managing Director
NCR	Non-Conformance Report
PO	Purchase Order
PROCESS	BOM in-house manufacture
PURCHASE	BOM external sourcing
QC	Quality Control
SO	Sales Order

C. Environment Variables (essential)

```
# Required
MY_SQL=localhost:3306
DB_NAME=bvc24
SECRET_KEY=<32+ random chars>
ALGORITHM=HS256

# Email – Resend (preferred)
RESEND_API_KEY=re...
SMTP_FROM=erp@bvc24.in
SMTP_FROM_NAME=BVC24 ERP

# Or SMTP fallback
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_USER=...
SMTP_PASSWORD=...
SMTP_USE_TLS=true

# Approver / URLs
APPROVER_EMAIL=md@bvc24.in
APPROVER_NAME=Managing Director
FRONTEND_URL=https://erp.bvc24.in
BACKEND_URL=https://erp.bvc24.in/api

# WhatsApp (optional, CallMeBot – free)
CALLEBOT_API_KEY=...
MD_WHATSAPP_NUMBER=+91...

# Or WhatsApp Cloud API
WHATSAPP_TOKEN=EAAQ...
WHATSAPP_PHONE_NUMBER_ID=...

# Gemini (optional)
GEMINI_API_KEY=AIza...
GEMINI_MODEL=gemini-1.5-flash

# Dev only
EMAIL_TESTING_OVERRIDE_TO=test@example.com
```

See [docs/appendix/C-environment-variables.md](#) for the complete reference.

D. Roadmap

Planned phases

- **Phase 6** — Invoicing with GST e-Invoice IRN, e-Way Bill, credit/debit notes, payment reconciliation.
- **Phase 7** — Service / AMC contracts, complaint tickets, service visit scheduling, spare-parts consumption, renewal reminders.

Cross-cutting

- Alembic migrations replacing the `IF NOT EXISTS` block when schema changes get more complex.
- Per-permission decorator enforcement across all routes.
- Refresh tokens to extend session.
- Rate limiting on login.
- Background jobs (Celery + Redis) for heavy reports and bulk emails.
- Materialised summary tables when dashboard load grows.
- Mobile app (React Native shell).
- Tamil + Hindi UI translations.
- Per-department approver configuration.
- WebSocket layer when polling becomes a bottleneck.
- Knowledge-base chatbot (RAG over this `/docs` folder).
- PDF attachment variant of customer-facing documents.
- Multi-currency support.
- Centralised audit log table.

Known Gaps in Release 1.0

- Seed endpoints not gated by environment.
- Permission enforcement is role-level, not per-permission.
- Refresh-token endpoint absent.
- No rate limiting on `/admin-login`.
- Email queue is synchronous.
- File uploads have no virus scanning.

E. Changelog

Release 1.0 — Production Ready (current)

- **Phase 5 — Sales Orders with Payment-Gated Workflow:** Awaiting_Advance status, advance amount + due date in email, auto-confirm on advance payment, MD WhatsApp alerts at three SO events, advance banner in UI.
- **Production & Gantt rewrite:** 30-day calendar header, one-stage-per-working-day, Sundays skipped, drawer 98vw, auto-assignment removed, sales-role employees excluded from production stages.
- **Sticky-header modal pattern** applied across Employee profile, Quotation, Sales Order, Purchase Order modals + Work Order Gantt drawer.

- **WhatsApp notifications:** `whatsapp_service.py` with CallMeBot + Cloud API, `notify_md_safe()`, wired across CRM and SO events.
- **Inventory rebuild:** BVC red theme, KPI tiles, MaterialCard with stock fill bar, AdjustModal, DetailDrawer, category auto-detection.
- **BVC red theme** applied consistently (login, sidebar, modals, chatbot, gradients).
- **Chatbot:** `gemini_service.py` (general ERP) + `hr_assistant.py` (stateless rule-based for leave).

Release 0.4 — Procurement (Phase 4)

Supplier master, PO CRUD with workflow, GRN with partial receipts and rejections, GRN-to-inventory pipeline, auto-PO from project BOM, PO email + rejection-notice resend.

Release 0.3 — CRM & Quotations (Phase 3)

Customer master with lead pipeline, contacts, requirements, Quotation CRUD with lines/totals/terms, SEND → APPROVE → CONVERT workflow, public share token, email dispatch with view telemetry, auto-pricing from BOM, A4 print.

Release 0.2 — HR Operations (Phase 2)

Biometric-driven attendance, leave with token-based email approval (CASUAL/SICK/EARNED quotas, auto-LOP), payroll runs from attendance + tasks, STAR performance with weighted formula and recommendations, MD Review.

Release 0.1 — Foundation (Phase 1)

Vendor / multi-tenancy, MANUFACTURING org preset, Employee master with UUID PK and profile self-registration, JWT auth (admin + employee), bcrypt passwords, auto-migration framework.

End of Document

For the multi-file modular edition see `docs/README.md` and the structured folders:

- `docs/01-executive-summary.md` ... `docs/06-database-schema.md` — Foundation
- `docs/modules/01-organization.md` ... `docs/modules/16-print-and-public-links.md` — 16 module documents
- `docs/api/01-auth.md` ... `docs/api/14-misc.md` — Complete API reference
- `docs/appendix/A-glossary.md` ... `docs/appendix/E-changelog.md` — Reference appendix